

Unit 302 Mock exam

Question 6: For a coil of 0.2H inductance, 35 ohms resistance, on 230V 50Hz supply, calculate: X_L , Z and Power factor. Draw an impedance triangle to scale.

First, calculate the X_L .

Step one write the base formula

Sometimes the Hertz value is not given. Therefore, assume that the Hz is 50. Whereas, in America it is 60 Hz.

$$X_L = 2 \times \pi \times \begin{matrix} F \\ \text{Hz} \end{matrix} \times \begin{matrix} L \\ \text{H or mH} \end{matrix}$$

Step two swap in the values

$$X_L = 2 \times \pi \times 50 \text{ Hz} \times 0.2 \text{ H}$$

Step three

Deduce the answer $X_L = 62.83\Omega$

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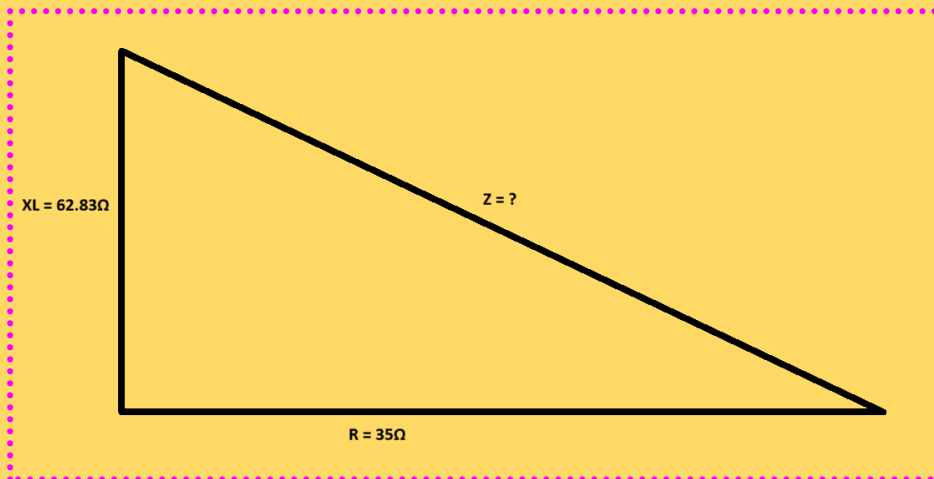
Second, calculate the Z value.

Z is also referred to as Impedance or Total resistance.

Draw the base formula for Z.

$$Z = \sqrt{R^2 + XL^2}$$

Draw the triangle representation as well.



XL is figured out in part one.
R is provided in the question.

$$Z = 71.92$$

Thirdly, calculate the power factor.

$$= PF \Rightarrow R / Z \Rightarrow 35 / 71.92 = 0.486$$

= $\cos^{-1}(0.486)$ To convert into degrees

$$= 60.92$$

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